

CLASH OF THE **CLAWS**

RV College of Engineering

TEAM SYNAPSE

Protecting focus in high-stress digital environments through intelligent on-device interventions.

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Problem Statement

- In today's digital world, people are constantly exposed to notifications, social media stimulation, multitasking, and continuous information overload.
- This has significantly reduced attention spans and increased stress, burnout, and mental fatigue among students and working professionals.
- Existing devices only respond to direct user input and are unable to understand the user's cognitive condition or stress level.
- We selected the On-Device AI and Productivity theme because we wanted to build a system that not only improves productivity but also actively protects the user's mental focus in real time.
- Our solution addresses the growing issue of cognitive overload by creating a system that can intelligently detect stress and intervene before the user becomes overwhelmed.

Current Solution & Gaps

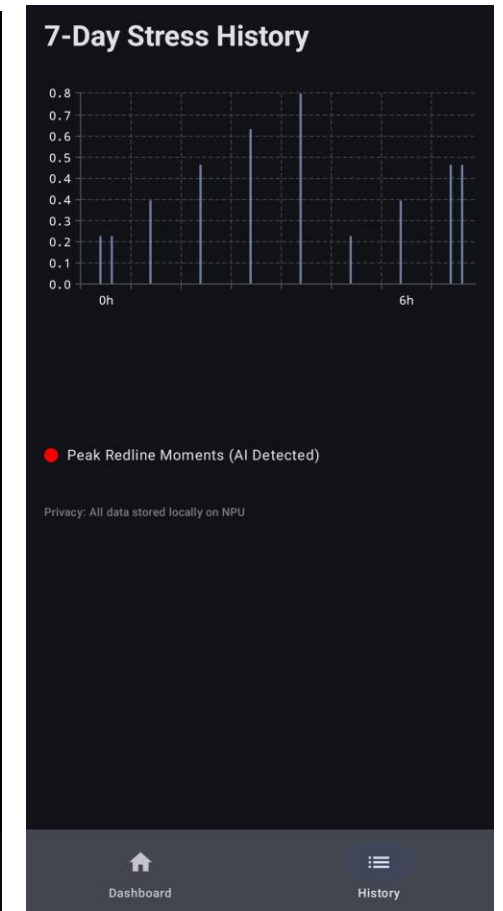
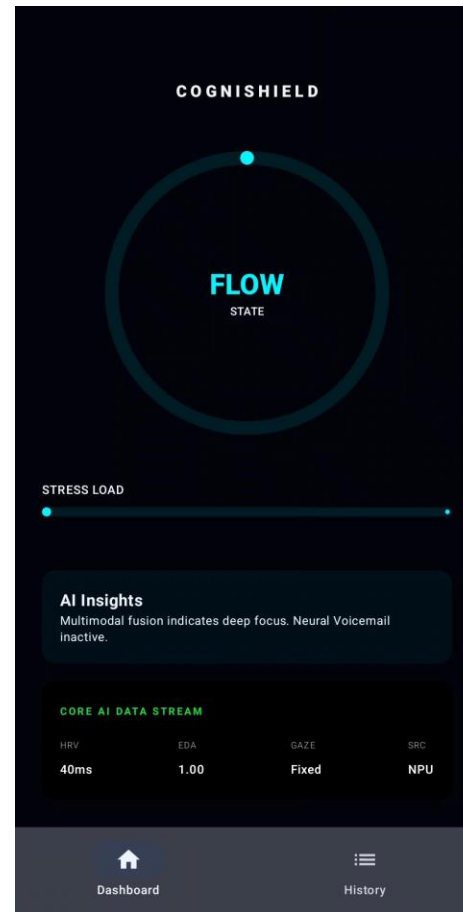
- Many existing applications such as focus timers, screen-time trackers, and productivity tools attempt to reduce distractions, but most of them rely heavily on manual activation.
- Applications like Focus Mode or Digital Wellbeing can mute notifications or restrict app usage, but they cannot understand when the user is mentally overloaded.
- These systems react only after the user manually enables them and do not adapt dynamically to changing stress conditions.
- There is currently a major gap in systems that can intelligently monitor cognitive state and automatically adjust device behavior in real time.
- This creates an opportunity for a smarter, AI-driven productivity system that can proactively protect users from burnout and excessive stimulation

Your Solutions

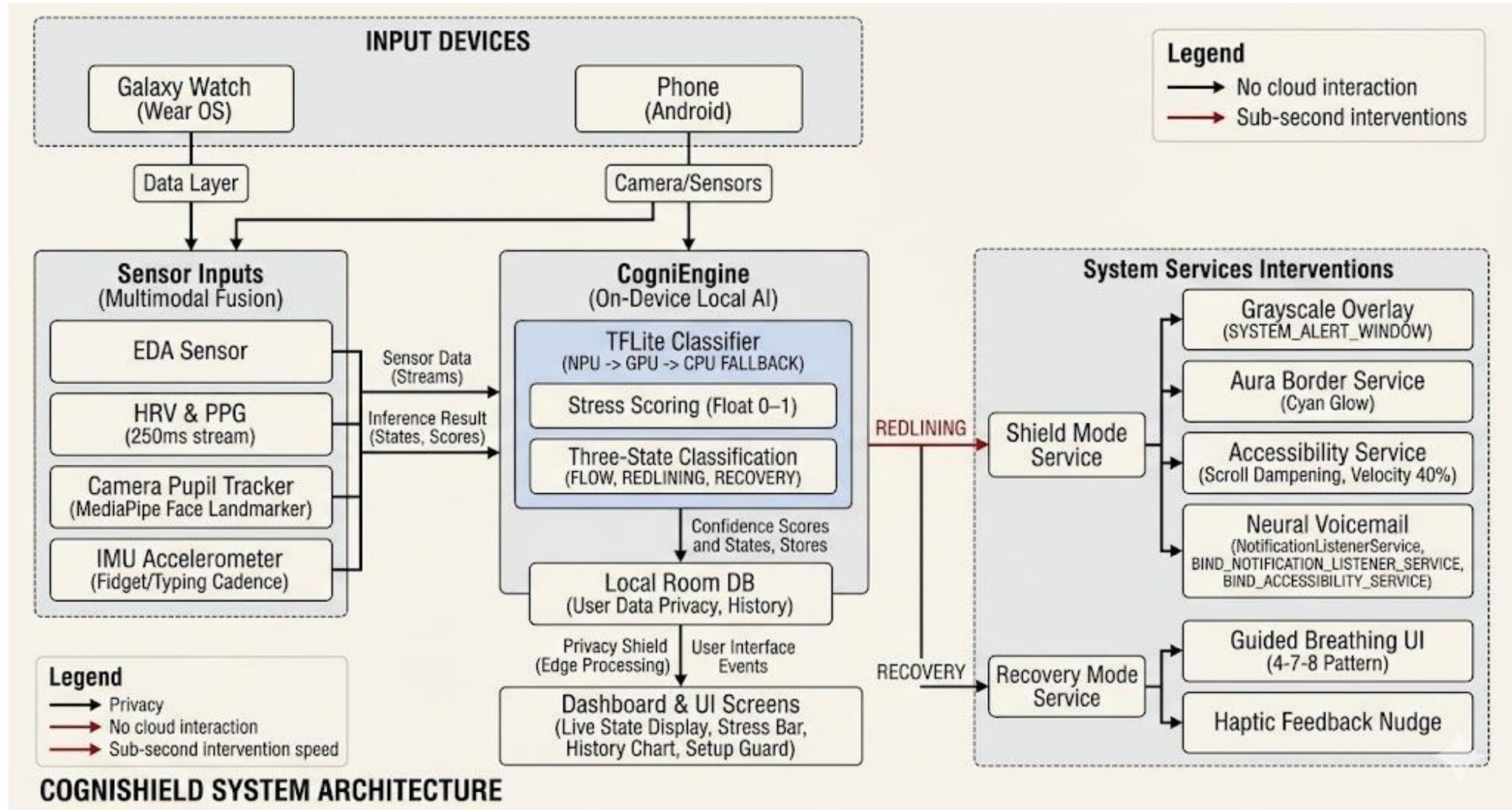
- CogniShield is a real-time cognitive intervention system that continuously monitors the user's cognitive state and dynamically changes the device experience to reduce stress and improve focus.
- The system operates using three cognitive states: FLOW, REDLINING, and RECOVERY.
- FLOW represents a calm and productive state, REDLINING indicates high cognitive stress or overload, and RECOVERY helps guide the user back to a balanced state.
- When the system detects overload, Shield Mode is automatically activated. This includes grayscale mode to reduce dopamine-triggering colors, scroll dampening to slow rapid content consumption, notification suppression to minimize interruptions, and a calming aura border around the screen.
- During the recovery phase, the app launches guided breathing exercises and haptic feedback to help the user relax and regain focus.
- The entire system is designed with a privacy-first approach where all processing occurs directly on the device

Demo/Product Walkthrough

The demonstration begins with the user in the FLOW state, where the device functions normally with a calm and responsive interface. When cognitive overload is detected, the system shifts to the REDLINING state and automatically activates Shield Mode by enabling grayscale mode, slowing scrolling, suppressing notifications, and displaying a glowing focus border. As the user relaxes, the system enters RECOVERY mode, launching breathing exercises and gentle haptic feedback to help restore focus and reduce stress.



Tech & Architecture



Impact & Use cases

- CogniShield has the potential to significantly improve productivity and mental wellness across multiple user groups.
- Students preparing for exams often experience high levels of digital distraction and stress, making it difficult to maintain concentration for long periods.
- Professionals working in fast-paced environments constantly face notification overload, multitasking pressure, and mental fatigue.
- Remote workers and content creators are also highly exposed to continuous digital stimulation throughout the day.
- CogniShield helps these users by intelligently reducing distractions during periods of cognitive overload and guiding them toward recovery when stress levels become too high. Beyond individual use, the system also has potential applications in enterprise productivity systems, wellness platforms, and wearable technology ecosystems.

Differentiation

- CogniShield can help students, professionals, developers, and remote workers manage digital stress and maintain focus.
- The system reduces overstimulation during high-pressure situations and encourages healthier technology usage through intelligent interventions.

Demo Link

Git Hub link: <https://github.com/khushi-n-murthy/CogniShield>

Demo link: <https://drive.google.com/drive/folders/1oH0NNQcF5-k0b0CjPgpgpwUzdrEW3jg?usp=sharing>

THANK YOU!